

A one-station counterexample to Conjecture 00000005508

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The clause being disproved

The original conjecture defines a bottleneck as the first station to saturate and asserts that the number of tied bottlenecks is the multiplicity of the minimum residual capacity, and is at most the logarithm of the station count. The Chinese version states the same bound on that multiplicity. There is no restriction excluding a network with one station. We disprove this bound, and therefore the conjunction, without needing to assess its identification or switching clauses.

Network and proof

Consider a network consisting of one station with service capacity $\mu_0 = 1$. Jobs enter at rate $t \geq 0$ and leave after service. Its traffic coefficient is $a_0 = 1$, its offered traffic is $ta_0 = t$, and its residual capacity is

$$r_0(t) = \mu_0 - ta_0 = 1 - t.$$

Saturation means that offered traffic reaches or exceeds capacity: $\mu_0 \leq ta_0$. The station is unsaturated for every $0 \leq t < 1$ and saturates at $t = 1$. It is therefore the first station to saturate. The bottleneck set is the singleton $\{0\}$ and has cardinality one. The residual-capacity vector has one entry, so its minimum has multiplicity one at every load, including the saturation load. We use precisely the multiplicity convention explicitly specified in the conjecture; it does not assign multiplicity zero to a unique minimizer.

The station count is $n = 1$. The bound would read

$$1 \leq \log 1 = 0,$$

which is false. This is independent of logarithm base: for every genuine base $b > 0$, $b \neq 1$, $\log_b 1 = 0$. For comparison, a network of two identical independent stations receiving rate t each has residual vector $(1 - t, 1 - t)$ and two tied bottlenecks at $t = 1$. It also violates the bound for natural logarithms ($\log 2 < 1 < 2$), base two ($\log_2 2 = 1 < 2$), and base ten ($\log_{10} 2 < 1 < 2$). The singleton witness additionally resolves every possible legitimate base, without relying on a plurality convention. The claim is an exact bound; interpreting it as an asymptotic order bound or adding an additive constant would change the conjecture.

Formal verification and scope

The Lean project uses Mathlib's real numbers and natural logarithm. **Network** `n` records positive capacities and positive traffic coefficients on the actual station type `Fin n`. **Saturated** is the capacity inequality above. **FirstBottleneck** requires a nonnegative load at which the station saturates and the absence of saturation at every earlier nonnegative load, for every station. The bottleneck set is the filtered finite set of stations satisfying this predicate, and its cardinality is computed from that set.

single_first proves first saturation at load one. **single_tied_count** proves that the defined bottleneck set has cardinality one. The minimum multiplicity is computed independently by the minimum-station set; the two counts agree at saturation. The final theorem **conjecture_5508_false** refutes the universally quantified logarithmic tie bound. **single_violates_every_base** verifies the base-change expression $\log 1 / \log b$ as well. The two-station comparison is supplementary; the fully formalized singleton witness alone disproves the exact original claim. No statistical assumption or queueing simulation is needed: this is a deterministic traffic/capacity network.